

Assignment - 1

1. Iterate from 0 to 200 and return te odd numbers.

```
for i in range(0, 200):
    if i % 2 != 0:
        print(i, end = " ")
```

1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 8

2. Write code using lambda function that takes three parameters and adds first two.

```
num1 = [1, 2, 3]
num2 = [4, 5, 6]
num3 = [7, 8, 9]

result = map(lambda n1, n2, n3 : n1 + n2, num1, num2, num3)
print(result)
print("Result : ",list(result))
```

<map object at 0x7efed4a2b7c0>
Result : [5, 7, 9]

3. Write your own program to show the use of lambda function.

```
first_name = ['Tony', 'Steve', 'Chris', 'Paul']
last_name = ['Stark', 'Rogers', 'Hemsworth', 'Bid']

res = map(lambda i, j : i + " " + j, first_name, last_name)

x = list(res)
for i in x:
    print(i)
```

Tony Stark
Steve Rogers
Chris Hemsworth
Paul Bid

4. Program to double each item in a list using map().

```
numbers = (1, 2, 3, 4, 5)
result = list(map(lambda x: x*2, numbers))
print("Result : ", result)
```

Result : [2, 4, 6, 8, 10]

5. Implementation of the quicksort algorithm in python [3, 6, 8, 10, 1, 2, 1].

```
def quicksort(arr):
    if len(arr) <= 1:
        return arr

    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]

    return quicksort(left) + middle + quicksort(right)

numbers = [3, 6, 8, 10, 1, 2, 1]
sorted_numbers = quicksort(numbers)
print("Sorted numbers : ", sorted_numbers)
```

Sorted numbers : [1, 1, 2, 3, 6, 8, 10]

6. Compute square of numbers using mapping function / lambda function.

```

numbers = [1, 2, 3, 4, 5]
squared = list(map(lambda n : n*n, numbers))

print("Square of numbers : ", squared)

Square of numbers :  [1, 4, 9, 16, 25]

```

7. Compute square of numbers using list comprehension technique.

```

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
result = [i*i for i in numbers]
print("Square of numbers : ", result)

Square of numbers :  [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]

```

8. Compute square of even numbers using list comprehension / set comprehension techniques.

```

#List comprehension

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
result = [i*i for i in numbers if i % 2 == 0]
print("Square of even numbers : ", result)

Square of even numbers :  [4, 16, 36, 64, 100]

```

```

#Set comprehension

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
result = {i*i for i in numbers if i % 2 == 0}
print("Square of even numbers : ", result)

Square of even numbers :  {64, 100, 4, 36, 16}

```

9. Explain the following code:

```

d = {'person' : 2, 'cat' : 4, 'spider' : 8}
for animal in d:
    legs = d[animal]
    print('A %s has %d legs' %(animal, legs))

```

Answer :

- The first line creates a dictionary named 'd', which contain :
 - keys : 'person', 'cat', 'spider'
 - values : 2, 4, 8
- These values are mapped to respective keys.
- In the second line, 'for' loop is used to iterate over dictionary 'd'.
- By default, it iterates through the keys.
- On the first loop, the variable 'animal' becomes 'person', then 'cat' and at last 'spider'.
- The line 'legs = d[animal]', uses the key (animal) to fetch it's matching values from the dictionary 'd'.
- For example, if animal = 'person', then 'd[animal]' = '2'.
- In the last line :
 - %s : placeholder for a string
 - %d : placeholder for a decimal integer
- It is going to print keys and respective values from the dictionary as follows :
- Output :

A person has 2 legs

A cat has 4 legs

A spider has 8 legs

